

Smart Campus AI System

Product Requirements Document (PRD)

Version: 1.0

Product Name: Smart Campus AI System

Product Type: Web-Based Campus Management Platform

1. Executive Summary

The Smart Campus AI System is an AI-enabled web platform designed to improve campus operations and student support services. The platform centralizes complaint management, student assistance, and administrative monitoring into a single system.

The product aims to reduce manual workload, improve communication efficiency, and provide transparency in issue resolution through automation and real-time tracking.

2. Problem Statement

Educational institutions often face challenges in managing student concerns effectively. Common issues include:

- Delayed responses to student queries
- Manual complaint handling processes
- Lack of complaint transparency
- Difficulty tracking issue resolution
- Limited visibility into operational performance

These challenges negatively impact student satisfaction and administrative efficiency.

The Smart Campus AI System addresses these issues through automation, AI assistance, and centralized management.

3. Product Vision

To create an intelligent campus ecosystem that enhances communication, improves operational efficiency, and provides seamless support services for students and administrative departments.

4. Product Goals

Primary Goals

- Provide instant AI-based assistance to students.
- Digitize complaint management processes.
- Improve issue resolution efficiency.
- Enable role-based administration.
- Deliver actionable analytics for decision-making.

Business Goals

- Reduce manual administrative workload.
- Increase student satisfaction.
- Improve complaint resolution transparency.
- Enable data-driven management decisions.

5. Stakeholders

Primary Stakeholders

- Students
- College Administration
- Head of Departments (HODs)
- Mentors
- Training and Placement Officers (TPOs)
- Accounts Department

Secondary Stakeholders

- College Management
- IT Support Team

6. User Personas

Student

Needs

- Quick answers to common questions
- Easy complaint submission
- Complaint status tracking

Pain Points

- Long response times
- Lack of issue visibility
- Manual processes

Administrative User

Needs

- Efficient complaint management
- Department-specific complaint access
- Complaint monitoring

Pain Points

- High manual workload
- Scattered complaint records
- Limited reporting capabilities

7. Functional Requirements

FR-01 User Authentication

The system shall provide secure login functionality for administrative users.

Acceptance Criteria

- Users must authenticate before dashboard access.
- Sessions must remain secure.

FR-02 Role-Based Authorization

The system shall restrict dashboard access according to assigned user roles.

Acceptance Criteria

- Admin can access all complaints.
- Department users can access relevant complaints only.

FR-03 AI Chatbot

The system shall provide an AI-powered chatbot for student support.

Supported Topics

- Fees
- Examinations
- Attendance
- Timetable

Acceptance Criteria

- Responses generated within three seconds.
- Query categories recognized correctly.

FR-04 Complaint Submission

The system shall allow students to submit complaints digitally.

Required Fields

- Student Name
- Email Address
- Complaint Category
- Complaint Description

Acceptance Criteria

- Complaint saved successfully.
- Unique Complaint ID generated.

FR-05 Complaint Tracking

The system shall allow students to track complaint progress.

Acceptance Criteria

- Complaint ID validation.
- Real-time status display.

FR-06 Complaint Management

The system shall enable authorized users to update complaint statuses.

Status Values

- Pending
- Seen
- Resolved

Acceptance Criteria

- Updates reflected immediately.
- Status history maintained.

FR-07 Delayed Complaint Detection

The system shall automatically identify unresolved complaints that exceed predefined resolution timelines.

Acceptance Criteria

- Delayed status assigned automatically.

FR-08 Analytics Dashboard

The system shall provide real-time complaint analytics.

Metrics

- Total Complaints
- Pending Complaints
- Seen Complaints
- Resolved Complaints

Acceptance Criteria

- Dashboard updates dynamically.

8. Non-Functional Requirements

Performance

- Application response time below 3 seconds.
- Dashboard loading time below 5 seconds.

Scalability

- Support future integration of additional modules.
- Handle increasing complaint volume efficiently.

Security

- Password-protected access.
- Role-based authorization.
- Secure session management.

Availability

- Target uptime of 99%.

Usability

- Mobile-responsive design.
- Intuitive user interface.
- Dark mode support.

Maintainability

- Modular architecture.
- Well-documented codebase.

9. System Architecture Overview

Frontend

- HTML
- CSS
- JavaScript

Backend

- Flask Framework
- Python

Database

- Relational Database Management System

Hosting

- Render Cloud Platform

Source Control

- GitHub

10. Assumptions

- Users have internet access.
- Administrative users possess authorized credentials.
- Complaint data is stored securely.
- Students understand complaint tracking procedures.

11. Constraints

- Initial chatbot functionality is rule-based.
- Mobile application is not included in Version 1.0.
- External communication channels are not integrated.

12. Risks

Risk	Impact	Mitigation
Incorrect chatbot responses	Medium	Continuous knowledge base updates
Unauthorized access	High	Strong authentication and role control
High complaint volume	Medium	Scalable backend architecture
Data loss	High	Regular backups

13. Product Roadmap

Phase 1 (Current Release)

- AI Chatbot
- Complaint Submission
- Complaint Tracking
- Role-Based Dashboard
- Analytics Dashboard

Phase 2

- Email Notifications
- Automated Complaint Assignment
- Department Performance Reports

Phase 3

- Generative AI Chatbot
- Mobile Application
- WhatsApp Integration
- Predictive Analytics
- Sentiment Analysis

14. Success Metrics

User Metrics

- Number of active students
- Complaint submission rate
- Chatbot usage frequency

Operational Metrics

- Average resolution time
- Resolution rate
- Delayed complaint percentage

Business Metrics

- Student satisfaction improvement
- Administrative workload reduction

Conclusion

The Smart Campus AI System is designed to modernize student support operations by combining AI assistance, complaint management, and administrative analytics into a unified platform. The system provides immediate value through automation and establishes a scalable foundation for future smart campus innovations.